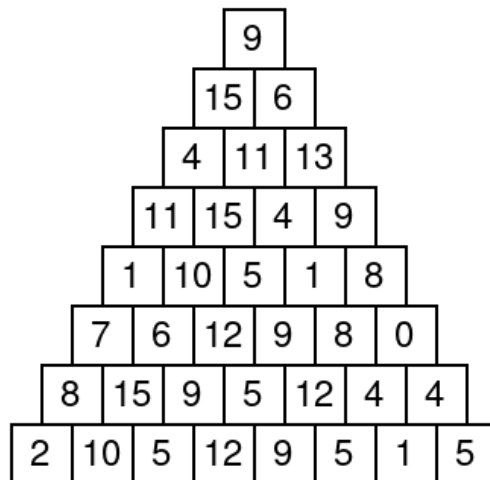


Xor Pyramid Row

Time limit: 1.00 s

Memory limit: 512 MB

Consider a xor pyramid where each number is the xor of lower-left and lower-right numbers. Here is an example pyramid:



Given the bottom row of the pyramid, your task is to find the numbers on the k -th row from the top.

Input

The first line has two integers n and k : the size of the pyramid and the given row.

The next line has n integers $a_1, a_2, \dots, a_n$: the bottom row of the pyramid.

Output

Print k integers: the numbers on the k -th row from the top.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq a_i \leq 10^9$

Example

Input	Output
8 5 2 10 5 12 9 5 1 5	1 10 5 1 8